

Ex. No:6:-

Use Sleuth Kit to analyze digital evidence

Date:-

Aim:-

To use The Sleuth Kit command-line tools to analyze a disk image and extract digital evidence.

Description:-

The Sleuth Kit (TSK) is a collection of command-line tools that allow you to analyze disk images and recover digital evidence. Here's a step-by-step guide to using Sleuth Kit on a Windows machine to analyze digital evidence:

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixc_xTU-y
- and download the latest version for Windows.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.

Step 2: Acquire the Disk Image

Before analysis, you need a disk image of the evidence. This can be an image of a hard drive, memory card, or any other storage device.

1. Create Disk Image:

- Use a tool like FTK Imager or dd to create a bit-by-bit copy of the storage device.
Ensure the image is in a format supported by Sleuth Kit, such as .dd, .raw, .img, or .E01.
- Download the below files from the google drive
 1. 4Dell Latitude CPi.E01
 2. 4Dell Latitude CPi.E02

Step 3: Mount the Disk Image (Optional)

Mounting the disk image makes it easier to analyze the file system.

1. Mount the Image:

- You can use a tool like OSFMount to mount the image as a virtual drive on your Windows system.
- This step is optional but helps with navigating the file system easily.

Step 4: Analyze the File System

Use Sleuth Kit tools to analyze the file system and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the Command Prompt and navigate to the directory where Sleuth Kit is installed.

2. Identify File System Type with fsstat:

- Run the command: `arduino fsstat [image file] > filesystem_info.txt`
- This command outputs information about the file system, which is crucial for understanding the structure of the disk.

3. List Partitions with mmls:

- Run the command:
arduino mmls [image file] > partitions.txt ○ This command lists the partitions within the image file.

4. Analyze File System with fls:

- Run the command: arduino
fls -r [image file] > file_list.txt ○ This command recursively lists files and directories in the file system, showing their metadata.

5. Recover Deleted Files with icat:

- To extract a specific file, use icat:
css icat [image file] [inode number] > [output file] ○ Replace [inode number] with the inode of the file you want to recover, which you can find from the fls output.

Step 5: Analyze Metadata

Extract metadata from files to understand more about the file's history.

1. View Metadata with istat:

- Run the command: css
istat [image file] [inode number] > metadata_info.txt ○ This provides detailed information about a file, including timestamps, size, and allocation status.

Step 6: Timeline Analysis (Optional)

Creating a timeline of file activity can be crucial in an investigation.

1. Create Timeline with mactime:

- Generate a body file using fls: css fls -m / -r [image file] > body.txt
- Then create the timeline: css
mactime -b body.txt > timeline.txt ○ The timeline includes MAC (Modified, Accessed, Changed) times of files.

Step 7: Generate a Report

Document your findings by compiling all the data into a comprehensive report.

1. **Compile the Data:** ○ Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).
2. **Analyze and Document:**
 - Review the findings, highlight important evidence, and write a report summarizing the investigation's results.

Step 8: Finalize and Store Evidence

Ensure that all evidence and reports are securely stored.

1. Archive Evidence:

- Use a secure method to archive the disk image and analysis results, ensuring integrity and confidentiality.

2. Store Securely:

- Store the archived data in a secure location, following the chain of custody procedures.

By following these steps, you can use Sleuth Kit on a Windows machine to effectively analyze digital evidence and extract crucial information for your investigation.

SCREENSHOTS:

```
C:\Users\nikhid>cd "%USERPROFILE%\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin"

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat
Missing image name
Usage: fsstat [-tvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-o ingoffset] image
-t: display type only
-i imgtype: The format of the image file (use '-i list' for supported types)
-b dev_sector_size: The size (in bytes) of the device sectors
-f fstype: File system type (use '-f list' for supported types)
-o ingoffset: The offset of the file system in the image (in sectors)
-P pooltype: Pool container type (use '-P list' for supported types)
-B pool_volume_block: Starting block (for pool volumes only)
-v: verbose output to stderr
-V: Print version
-k password: Decryption password for encrypted volumes

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>dir "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp"
Volume in drive C is OS
Volume Serial Number is 462C-EA9E

Directory of C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp

27-08-2026 00:51 <DIR> .
27-08-2026 00:50 <DIR> ..
26-08-2026 22:24 671,694,597 40ell Latitude Cpi.E01
26-08-2026 22:22 419,384,951 40ell Latitude Cpi.E02
                2 File(s) 1,090,479,548 bytes
                2 Dir(s) 70,902,882,304 bytes free

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\40ell Latitude Cpi.E01" > "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\filesystem_info.txt"
Cannot determine file system type

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>type "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\filesystem_info.txt"

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\40ell Latitude Cpi.E01" > "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\partitions.txt"
C:\Users\nikhid\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>type "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\partitions.txt"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot      Start      End      Length      Description
#00:  Meta      000000000000  000000000000  0000000001  Primary Table (#0)
#01:  -----      000000000000  000000000062  0000000063  Unallocated
#02:  000:000  00000000063  0009510479  0009510417  NTFS / exFAT (0x07)
#03:  -----      0009510408  0009514259  0000003780  Unallocated

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -r "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\40ell Latitude Cpi.E01" > "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\file_list.txt"
Cannot determine file system type

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>type "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\file_list.txt"

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>dir "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp"
Volume in drive C is OS
Volume Serial Number is 462C-EA9E

Directory of C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp

27-08-2026 00:56 <DIR> .
27-08-2026 00:50 <DIR> ..
26-08-2026 22:24 671,694,597 40ell Latitude Cpi.E01
26-08-2026 22:22 419,384,951 40ell Latitude Cpi.E02
27-08-2026 00:54 0 filesystem_info.txt
27-08-2026 00:56 0 file_list.txt
27-08-2026 00:55 427 partitions.txt
                5 File(s) 1,090,479,975 bytes
                2 Dir(s) 70,893,297,604 bytes free

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\40ell Latitude Cpi.E01" > "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\filesystem_info.txt"
Cannot determine file system type

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\40ell Latitude Cpi.E01" > "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\partitions.txt"

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```

```
++ r/r 3923-128-4: cga40852.fon
++ r/r 3470-128-3: ega40850.fon
++ r/r 3475-128-3: mvboli.ttf
++ r/r 3412-128-3: smalle.fon
++ r/r 3455-128-3: timesbd.ttf
++ r/r 4000-128-4: vga869.fon
++ r/r 3965-128-4: vgaf1257.fon
++ r/r 3403-128-3: vgafix.fon
++ r/r 3936-128-4: vgafixe.fon
++ r/r 4040-128-4: vgafixr.fon
++ r/r 4067-128-4: vgafixt.fon
++ r/r 4041-128-4: vgasysr.fon
++ r/r 4068-128-4: vgasyst.fon
+ r/r 5028-128-4: Gone Fishing.bmp
+ r/r 5029-128-4: Greenstone.bmp
+ d/d 9846-144-6: Installer
++ r/r 10179-128-3: ac704.msi
++ r/r 9860-128-3: d2a9f.msi
++ d/d 9861-144-1: {350C97B0-3D7C-4EE8-BAA9-00BCB3D54227}
++ r/r 9862-128-3: places.exe
++ d/d 10180-144-6: {6C31E111-96BB-4ADC-9C81-E6D3EE0DD8D3}
+++ r/r 10183-128-4: ARPPRODUCTICON.exe
+++ r/r 10181-128-3: htmlgen.exe
+++ r/r 10182-128-4: PowerCalc.exe
+++ r/r 10185-128-3: Readme
+++ r/r 10184-128-4: Timershot.exe
+++ r/r 10186-128-3: TweakUI.exe
+ r/r 11772-128-3: iun6002.exe
+ r/r 11734-128-4: Look@LAN Setup Log.txt
+ r/r 3651-128-3: explorer.scf
+ r/r 3653-128-3: msdfmap.ini
+ r/r 339-128-3: REGLOCS.OLD
+ r/r 3680-128-1: setuperr.log
+ r/r 5464-128-1: vb.ini
+ r/r 3901-128-4: regopt.log
+ r/r 431-128-3: SchedLgU.Txt
+ r/- * 0: ~GLC0000.TMP
+ r/- * 0: ~GLH0000.TMP
+ r/- * 0: ~GLH0001.TMP
//V 12305: $OrphanFiles
r/r 343-128-1: hiberfil.sys
r/r 228-128-1: pagefile.sys

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -o 63 -r "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\40ell Latitude Cpi.E01" > "C:\Users\nikhid\OneDrive\Desktop\sleuth kit exp\file_list.txt"

C:\Users\nikhid>sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>
```



```

C:\Users\nikhi\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>type "C:\Users\nikhi\OneDrive\Desktop\sleuth kit exp\Greenstone_metadata.txt"
MFT Entry Header Values:
Entry: 5029      Sequence: 2
LogFile Sequence Number: 12931864
Allocated File
Links: 2

STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 0 ()
Created: 2004-08-20 03:54:22.071406400 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified: 2004-08-20 03:54:22.091435200 (India Standard Time)
Accessed: 2004-08-20 03:54:22.081420800 (India Standard Time)

FILE_NAME Attribute Values:
Flags: Archive
Name: GREENS-1.BMP
Parent MFT Entry: 458      Sequence: 458
Allocated Size: 26624      Actual Size: 26582
Created: 2004-08-20 03:54:22.071406400 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified: 2004-08-20 03:54:22.081420800 (India Standard Time)
Accessed: 2004-08-20 03:54:22.081420800 (India Standard Time)

FILE_NAME Attribute Values:
Flags: Archive
Name: Greenstone.bmp
Parent MFT Entry: 458      Sequence: 458
Allocated Size: 26624      Actual Size: 26582
Created: 2004-08-20 03:54:22.071406400 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified: 2004-08-20 03:54:22.081420800 (India Standard Time)
Accessed: 2004-08-20 03:54:22.081420800 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-6) Name: N/A Resident size: 90
Type: $FILE_NAME (48-5) Name: N/A Resident size: 94
Type: $SECURITY_DESCRIPTOR (80-3) Name: N/A Resident size: 124
Type: $DATA (128-4) Name: N/A Non-Resident size: 26582 init_size: 26582
328204 328205 328206 328207 328208 328209 328210 328211
328212 328213 328214 328215 328216 328217 328218 328219
328220 328221 328222 328223 328224 328225 328226 328227

```

Rubrics:-

Criteria & Marks Assigned	Mark Allotted	Mark Awarded
1. GitHub Activity & Submission Regularity	3	
2. Application of Forensic Tools & Practical Execution	3	
3. Documentation & Reporting	2	
4. Engagement, Problem-Solving & Team Collaboration	2	
Total	10	

Result:-

The disk image was analyzed with mmls, fls, istat, icat, and mactime. The partitions and files were listed, a file's metadata and timestamps were extracted, and a timeline of file activity was generated.

